Introduction to Deep Equilibrium Nets (DEQNs)

(A Mini-Course, March 2026)

Galo Nuño (Banco de España) **Simon Scheidegger** (HEC, University of Lausanne)

Purpose of the Lectures

This self-contained mini-course introduces **Deep Equilibrium Nets (DEQNs)**, a method that uses deep neural networks to solve high-dimensional dynamic stochastic economic models. DEQNs replace traditional grid-based collocation with neural network training, where the loss function is derived directly from the economic equilibrium conditions — no labeled data is needed.

The course covers the necessary ML/DL foundations, develops the DEQN methodology from scratch, applies it to the Brock–Mirman benchmark (which has an analytical solution), introduces practical tools such as Neural Architecture Search and loss normalization (ReLoBRaLo), and scales up to the multi-country International Real Business Cycle (IRBC) model.

The lectures combine theoretical discussions with hands-on coding exercises in Python.

Prerequisites

- Basic knowledge of macroeconomics (Euler equations, dynamic programming)
- Familiarity with Python and Jupyter notebooks (see the Python refresher included in this repository, or the more comprehensive QuantEcon Python Programming course)
- Linear algebra and calculus fundamentals (see Mathematics for Machine Learning)

Useful Background References

- **I. Goodfellow, Y. Bengio, A. Courville** (2016). *Deep Learning.* MIT Press. https://www.deeplearningbook.org
- **K. P. Murphy** (2022). *Probabilistic Machine Learning: An Introduction.* MIT Press.
- **G. James et al.** (2021). *An Introduction to Statistical Learning (ISLR).* Springer. https://www.statlearning.com

---

Software Requirements

All notebooks run on **Python 3.9+**. The main dependencies are:

Package	Version	Purpose	Used in
NumPy	>= 1.24	Numerical computing	All notebooks
SciPy	>= 1.10	Scientific computing (optimization, linear algebra)	06, 07
Pandas	>= 2.0	Data analysis	05
Matplotlib	>= 3.7	Visualization	All notebooks
scikit-learn	>= 1.3	Gaussian Process regression, kernels	07
TensorFlow	>= 2.15	Deep learning (DEQNs, surrogates)	01–05

Package	Version	Purpose	Used in
[TensorFlow Probability]	>= 0.23	Probabilistic modeling (stochastic shocks)	02
[PyTorch]	>= 2.0	Deep learning (surrogate primer)	06

Installation

We recommend using a virtual environment:

```
# Create and activate a virtual environment
python -m venv .venv
source .venv/bin/activate    # Linux/macOS
# .venv\Scripts\activate     # Windows

# Install all dependencies
pip install -r requirements.txt
```

Note: TensorFlow and PyTorch are both required since different notebooks use different frameworks (notebooks 01–05 use TensorFlow/Keras; notebook 06 uses PyTorch).

Course Overview

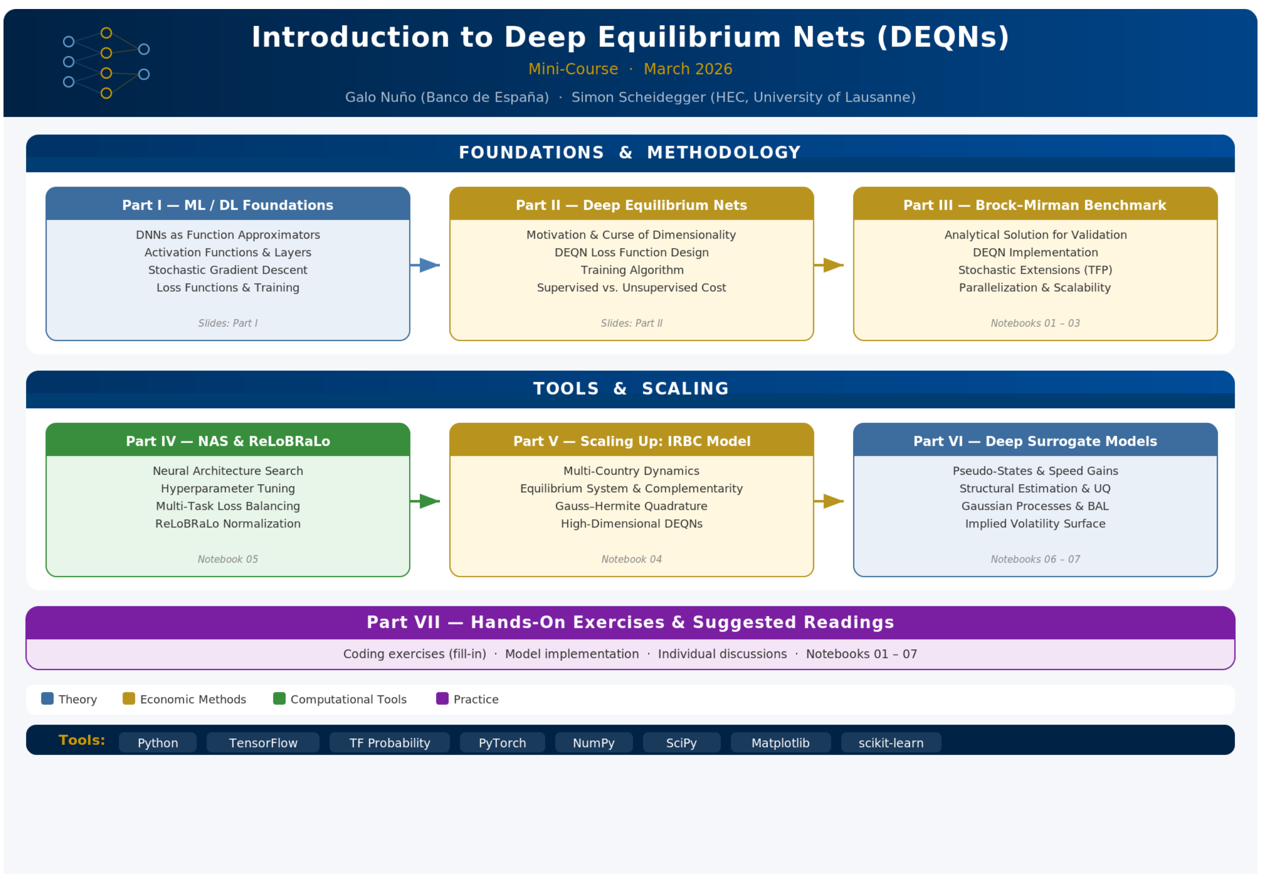

Figure 1: Course Overview

Course Content

The slide deck (DEQN_MiniCourse.tex) is organized as follows:

Part	Topic	Content
I	ML/DL Foundations	DNNs as function approximators, activation functions, SGD, loss functions
II	Deep Equilibrium Nets	Motivation, curse of dimensionality, DEQN loss function, training algorithm
III	Brock–Mirman Benchmark	Analytical test case, DEQN implementation, parallelization & scalability
IV	Practical Tools	Neural Architecture Search (NAS) and ReLoBRaLo loss normalization
V	Scaling Up — The IRBC Model	Multi-country model, equilibrium system, high-dimensional DEQNs
VI	Deep Surrogate Models	Pseudo-states, speed gains, structural estimation, UQ, GPs, Bayesian Active Learning
VII	Hands-on Exercises	Notebook overview and suggested readings

---

Code Notebooks

#	Notebook	Topic
01	01_Brock_Mirman_1972_DEQN	DEQN: Deterministic Brock–Mirman with analytical benchmark
02	02_Brock_Mirman_Uncertainty_DEQN	DEQN: Stochastic Brock–Mirman with TFP shocks
03	03_DEQN_Exercises_Blancs	DEQN Exercises (blanks to fill in)
03s	03_DEQN_Exercises_Solutions	DEQN Exercises (solutions)
04	04_IRBC_DEQN	DEQN: International Real Business Cycle Model
05	05_Neural_Architecture_Search	NAS via random search: hyperparameter tuning for DNNs
06	06_Surrogate_Primer	Deep surrogates: Black-Scholes pricing & implied volatility inversion
07	07_GP_and_BAL	GP regression from scratch, scikit-learn, Bayesian Active Learning

---

Readings

Paper	File
Azinovic, M., Gaegauf, L., & Scheidegger, S. (2022). Deep Equilibrium Nets. *International Economic Review*, 63(4), 1471–1525.	Azinovic_Gaegauf_Scheidegger_2022_DEQN.pdf
Fernández-Villaverde, J., Nuño, G., & Perla, J. (2024). Taming the Curse of Dimensionality: Quantitative Economics with Deep Learning. *NBER Working Paper 33117*.	taming.pdf
Chen, H., Didisheim, A., & Scheidegger, S. (2026). Deep Surrogates for Finance: With an Application to Option Pricing. *Journal of Financial Economics*, 177, 104222.	DeepSurrogates_JFE.pdf
Friedl, A., Kübler, F., Scheidegger, S., & Usui, T. (2023). Deep Uncertainty Quantification: With an Application to Integrated Assessment Models.	DeepUQ_with_an_application_to_IAM.pdf

---

Citation

If you find this course material useful for your research, please cite the following papers:

```
@article{azinovic2022deep,
  title={Deep Equilibrium Nets},
  author={Azinovic, Marina and Gaegauf, Luca and Scheidegger, Simon},
  journal={International Economic Review},
  volume={63},
  number={4},
  pages={1471--1525},
  year={2022},
  publisher={Wiley},
  doi={10.1111/iere.12575}
}

@techreport{fernandezvillaverde2024taming,
  title={Taming the Curse of Dimensionality: Quantitative Economics with Deep Learning},
  author={Fern{\'a}ndez-Villaverde, Jes{\'u}s and Nu{\~n}o, Galo and Perla, Jesse},
  year={2024},
  institution={National Bureau of Economic Research},
  type={Working Paper},
  number={33117}
}
```

License